Homework 2

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Problem 1

We use proof by induction. For the case of $n = 1$, we trivially have

$$\bigcup_{i=1}^{1} \overline{A}_i = \overline{A}_1 = \overline{\bigcup_{i=1}^{1} A_i}.$$

I will also show the case for $n = 2$ as a lemma to the inductive step. Let $x \in \overline{A_1 \cup A_2}$. It follows that for all closed sets K containing $A_1 \cup A_2$, we have $x \in K$. Note that since $A_1 \subseteq \overline{A}_1$ and $A_2 \subseteq \overline{A}_2$, we have $A_1 \cup A_2 \subseteq \overline{A}_1 \cup \overline{A}_2$ fairly trivially. Since $\overline{A}_1$ and $\overline{A}_2$ are closed, and the union $\overline{A}_1 \cup \overline{A}_2$ is finite, we have that $\overline{A}_1 \cup \overline{A}_2$ is closed. Therefore, $\overline{A}_1 \cup \overline{A}_2$ is a closed set containing $\overline{A_1 \cup A_2}$, and $x \in \overline{A}_1 \cup \overline{A}_2$. Therefore, $\overline{A_1 \cup A_2} \subseteq \overline{A}_1 \cup \overline{A}_2$.

Now the other direction. Let $x \in \overline{A}_1 \cup \overline{A}_2$. It follows that $x \in \overline{A}_1$ or $x \in \overline{A}_2$. Without loss of generality, assume $x \in \overline{A}_1$. Note that $A_1 \subseteq A_1 \cup A_2 \subseteq \overline{A_1 \cup A_2}$, and thus $x \in \overline{A_1 \cup A_2}$. Therefore, $\overline{A}_1 \cup \overline{A}_2 = \overline{A_1 \cup A_2}$.

Now for the inductive case. Let the statement hold for some $n > 1$, and consider $n+1$. We have

$$\overline{\bigcup_{i=1}^{n+1} A_i} = \overline{\left(\bigcup_{i=1}^{n} A_i\right) \cup A_{n+1}}.$$

Since $\bigcup_{i=1}^{n} A_i$ and A_{n+1} are just subsets of X, this is the closure of the union of two subsets of X. By the previous lemma, we have that

$$\overline{\left(\bigcup_{i=1}^{n} A_i\right) \cup A_{n+1}} = \left(\overline{\bigcup_{i=1}^{n} A_i}\right) \cup \overline{A}_{n+1}.$$

By the inductive hypothesis, we have

$$\left(\overline{\bigcup_{i=1}^{n} A_i}\right) \cup \overline{A}_{n+1} = \left(\bigcup_{i=1}^{n} \overline{A}_i\right) \cup \overline{A}_{n+1} = \bigcup_{i=1}^{n+1} \overline{A}_i.$$

By induction, the statement is proven.

For a counter example, consider —

Problem 2

- Clearly $\mathrm{Int}(A) \cup \mathrm{Int}(B) \subseteq \mathrm{Int}(A \cup B)$, but the converse does not hold in general. Consider a space X with nonempty subsets A, B, U such that $U \subsetneq A \cup B$, but $U \cap A \neq U$ and $U \cap B \neq U$. That is, U is partially contained by A and B individually, and fully contained by their union.

Now consider the topology $\mathcal{T} = \{\varnothing, U, X\}$ on X. Note that this set is closed under union and intersection:

$$U \cup \varnothing = U, \qquad U \cup X = X,$$

$$U \cap \varnothing = \varnothing, \qquad U \cap X = U.$$

Therefore, $\mathcal{T}$ is a topology on X. Also note that

$$\mathrm{Int}(A \cup B) = \varnothing \cup U = U,$$

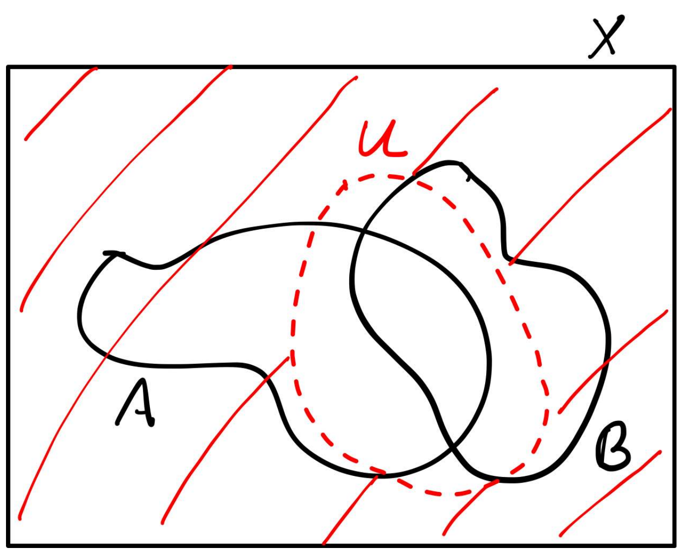
but since U is not a subset of A or B, we have that $\mathrm{Int}(A) = \mathrm{Int}(B) = \varnothing$. Since $\varnothing \cup \varnothing = \varnothing \neq U$, the statement does not hold by counterexample.

- This statement holds. Note that $\mathrm{Int}(A) \subseteq A$ and $\mathrm{Int}(B) \subseteq B$, so $\mathrm{Int}(A) \cap \mathrm{Int}(B) \subseteq A \cap B$. Also note that since $\mathrm{Int}(A), \mathrm{Int}(B)$ are open, this is a finite intersection of open sets, so $\mathrm{Int}(A) \cap \mathrm{Int}(B)$ is open. Since $\mathrm{Int}(A) \cap \mathrm{Int}(B) \subseteq A \cap B$, it follows that $\mathrm{Int}(A) \cap \mathrm{Int}(B) \subseteq \mathrm{Int}(A \cap B)$.

  Now the other direction. Let $x \in \mathrm{Int}(A \cap B)$. It follows that there exists some open set $U \subseteq A \cap B$ containing x. It follows that $U \subseteq A$ and $U \subseteq B$, and since U is open, we have $U \subseteq \mathrm{Int}(A), \mathrm{Int}(B)$. Therefore, $U \subseteq \mathrm{Int}(A) \cap \mathrm{Int}(B)$, and thus $x \in \mathrm{Int}(A) \cap \mathrm{Int}(B)$. Therefore, $\mathrm{Int}(A \cap B) \subseteq \mathrm{Int}(A) \cap \mathrm{Int}(B)$, and we are done.

- It's easily shown $\overline{A} \cap \overline{B} \subseteq \overline{A \cap B}$, but this statement fails in the other direction, by similar logic to the first statement. Let $A, B, U \subsetneq X$ be nonempty sets such that U and $A \cap B$ are disjoint, but U and A are not disjoint, and U and B are not disjoint. Once again $\mathcal{T} = \{\varnothing, U, X\}$ is a topology by the same argument as in statement 1. Consider the closed set U^c, which I will denote V for simplicity. Since U and $A \cap B$ are disjoint, this implies that $A \cap B \subseteq X \setminus U = V$, and thus $V \cap (A \cap B) = A \cap B$. Since V is the only closed set other than $\varnothing$ and X, and $V \subsetneq X$, we have that $\overline{A \cap B} = V$, since this is the smallest closed set containing $A \cap B$. However, since U and A are not disjoint, we have that $U \cap A \neq \varnothing$, and V does not contain $U \cap A$. Therefore, $V \cap A \neq A$, and thus A is not a subset of V. By the same logic, B is not a subset of V either. Therefore, $\overline{A} = X$ and $\overline{B} = X$, and since $X \cap X = X \neq V$, we have that $\overline{A \cap B} \neq \overline{A} \cap \overline{B}$.

  Here's a helpful image to visualize the argument I'm making.



Problem 3

Let A be closed. Then $A = \overline{A}$. Note that $\overline{A} \setminus \operatorname{Int}(A) \subseteq \overline{A}$, and $\partial A = \overline{A} \setminus \operatorname{Int}(A)$. Therefore, $\partial A \subseteq \overline{A} = A$.

Now for the converse, let $\partial A \subseteq A$. It follows that $\overline{A} \setminus \operatorname{Int}(A) \subseteq A$. Note that $\overline{A} = \operatorname{Int}(A) \cup (\overline{A} \setminus \operatorname{Int}(A))$. Since $\operatorname{Int}(A) \subseteq A$ and $\overline{A} \setminus \operatorname{Int}(A) = \partial A \subseteq A$, we have that their union must also be a subset of A. That is,

$$\overline{A} = \operatorname{Int}(A) \cup \big(\overline{A} \setminus \operatorname{Int}(A)\big) \subseteq A.$$

Since we also know for all sets that $A \subseteq \overline{A}$, we have that $A = \overline{A}$, and A is closed.

Problem 4

First we prove a brief lemma: for any function $f : X \to X$ for X any set, and for any collection $\{U_i\}_{i \in I}$ of subsets of X,

$$\bigcup_{i \in I} f^{-1}(U_i) = f^{-1}\left(\bigcup_{i \in I} U_i\right).$$

Let $x \in \bigcup_{i \in I} f^{-1}(U_i)$. It follows that there exists $i \in I$ such that $x \in f^{-1}(U_i)$, and thus $f(x) \in U_i$. Since $U_i \subseteq \bigcup_{i \in I} U_i$, we have that $f(x) \in \bigcup_{i \in I} U_i$, and thus

$$x \in f^{-1}\left(\bigcup_{i \in I} U_i\right).$$

Therefore, $\bigcup_{i \in I} f^{-1}(U_i) \subseteq f^{-1}\left(\bigcup_{i \in I} U_i\right)$. Now for the converse, let $x \in f^{-1}\left(\bigcup_{i \in I} U_i\right)$. It follows that $f(x) \in \bigcup_{i \in I} U_i$, and thus there exists $i \in I$ such that $f(x) \in U_i$. Therefore, $x \in f^{-1}(U_i)$, and since $f^{-1}(U_i) \subseteq \bigcup_{i \in I} f^{-1}(U_i)$, we have

$$x \in \bigcup_{i \in I} f^{-1}(U_i).$$

This proves the lemma.

Problem 5

Let $f : X \to Y$ be continuous, and let Y be Hausdorff.

Problem 6

Problem 7

I could not solve this without requiring that f be continuous, and in fact I was able to construct what I belivee is a counterexample if we assumme f is not continuous. I believe it was implied that f was supposed to be continuous, and have done the problem under this assumption. I only used this fact for a single statement at the end of the proof.

Let $\pi : \operatorname{Graph}(f) \to X$ be the canonical projection $(x, f(x)) \mapsto x$, and consider the function $\pi^{-1} : X \to \operatorname{Graph}(f)$ defined by $x \mapsto (x, f(x))$. Note that this map is well-defined because f is well defined; that is, if $x = y$, then we must have $f(x) = f(y)$ because f is a function, and thus $\pi^{-1}(x) = (x, f(x)) = (y, f(y)) = \pi^{-1}(y)$. Also note that this is clearly an inverse to π:

$$\left(\pi^{-1}\pi\right)(x, f(x)) = \pi^{-1}(x) = (x, f(x));$$

$$\left(\pi\pi^{-1}\right)(x) = \pi(x, f(x)) = x.$$

Therefore, π is a bijection. Now we show π is a homeomorphism. First, we show π is continuous. Let $U \subseteq X$ be open in X. Note that

$$\pi^{-1}(U) = \{(x, f(x)) \in \mathrm{Graph}(f) \mid x \in U\}.$$

I claim that $\pi^{-1}(X) = \mathrm{Graph}(f) \cap (U \times Y)$. Note that since $\mathrm{Graph}(f)$ inherits the subspace topology from the product topology on $X \times Y$, this is an open set. We have that $V_x \times Y$ is open in $X \times Y$, since V is open in X and Y is open in Y. It follows that $\mathrm{Graph}(f) \cap (U \times Y)$ is open in $\mathrm{Graph}(f)$ by definition of the subspace topology.

Now to show these sets are equal. Let $(x, f(x)) \in \pi^{-1}(U)$. It follows that $x \in U$, and since $f(x) \in Y$, we have $(x, f(x)) \in U \times Y$. Since we also have $(x, f(x)) \in \mathrm{Graph}(f)$, we have $(x, f(x)) \in \mathrm{Graph}(f) \cap (U \times Y)$. Therefore, $\pi^{-1}(U) \subseteq \mathrm{Graph}(f) \cap (U, Y)$.

Now the converse. Let $(x, f(x)) \in \mathrm{Graph}(f) \cap (U \times Y)$. It follows that $x \in U$, and thus $\pi(x, f(x)) = x \in U$. Therefore, $(x, f(x)) \in \pi^{-1}(U)$. Therefore, $\mathrm{Graph}(f) \cap (U \times Y) = \pi^{-1}(U)$.

Now, we show π^{-1} is continuous. Note that $\left(\pi^{-1}\right)^{-1} = \pi$, so we must show $\pi(U)$ is open for all open sets U. Let $U \subseteq \mathrm{Graph}(f)$ be open. Note that since $\mathrm{Graph}(f)$ is imbued with the subspace topology inherited from the product topology on $X \times Y$. That is, every open set U can be represented as a union

$$\bigcup_{i \in I} \mathrm{Graph}(f) \cap (V_i \times W_i),$$

for $\{V_i\}_{i \in I}$ open in X and $\{W_i\}_{i \in I}$ open in Y. By the same proof given in problem 4,

$$\pi\left(\bigcup_{i \in I} \mathrm{Graph}(f) \cap (V_i \times W_i)\right) = \bigcup_{i \in I} \pi\left(\mathrm{Graph}(f) \cap (V_i \times W_i)\right).$$

It then suffices to show $\pi\left(\mathrm{Graph}(f) \cap (V_i \times W_i)\right)$ is open in X for some fixed $i \in I$. Let $A \coloneqq \mathrm{Graph}(f) \cap (V \times W)$ for V, W open in X, Y, respectively. First, I will show that $\pi(A) = f^{-1}(W) \cap V$. Let $x \in \pi(A)$. It follows that $x \in V$ and $f(x) \in W$, and thus $x \in f^{-1}(W) \cap V$. Now let $x \in f^{-1}(W) \cap V$. It follows that $x \in V$ and $f(x) \in W$, and thus $(x, f(x)) \in V \times W$. Therefore, $(x, f(x)) \in A$, and $\pi(x, f(x)) = x \in \pi(A)$. Therefore, $\pi(A) = f^{-1}(W) \cap V$. Since f is continuous, $f^{-1}(W)$ is open in X, and by definition so is V. This is then a finite intersection of open sets, which is open. Therefore, $\pi(A)$ is open, and π^{-1} is continuous. Thus, π is a homeomorphism from $\mathrm{Graph}(f)$ to X.